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Introduction
to Artificial
Intelligence

Second Edition

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Chapter 1

First-order Predicate Logic

Many practical, relevant problems cannot be or can only very inconveniently be formulated in the language of propositional logic, as we can easily recognize in the following example. The statement

“Robot 7 is situated at the xy position (35, 79)”

can in fact be directly used as the propositional logic variable

`"Robot_7_is_situated_at_xy_position_(35, 79)"`

for reasoning with propositional logic, but reasoning with this kind of proposition is very inconvenient. Assume 100 of these robots can stop anywhere on a grid of 100×100 points. To describe every position of every robot, we would need $100 \cdot 100 \cdot 100 = 1000000 = 10^6$ different variables. The definition of relationships between objects (here robots) becomes truly difficult. The relation

“Robot A is to the right of robot B .”

is semantically nothing more than a set of pairs. Of the 10000 possible pairs of x -coordinates there are $(99 \cdot 98)/2 = 4851$ ordered pairs. Together with all 10000 combinations of possible y -values for both robots, there are $(100 \cdot 99) = 9900$ formulas of the type

`Robot_7_is_to_the_right_of_robot_12  $\Leftrightarrow$  Robot_7_is_situated_at_xy_position_(35, 79)  $\wedge$  Robot_12_is_sit`

defining these relations, each of them with $(10^4)^2 \cdot 0.485 = 0.485 \cdot 10^8$ alternatives on the right side. In first-order predicate logic, we can define a predicate

`Position(number, xPosition, yPosition)`. The above relation must no longer be enumerated as a huge number of pairs, rather it is described abstractly with a rule of the form

$$\forall u \forall v \text{is_further_right}(u, v) \Leftrightarrow \exists x_u \exists y_u \exists x_v \exists y_v \text{position}(u, x_u, y_u) \wedge \text{position}(v, x_v, y_v) \wedge x_u > x_v,$$

Where $\forall u$ is read as “for every u ” and $\exists v$ as “there exists v ”. In this chapter we will define the syntax and semantics of first-order predicate logic (PL1), show that many applications can be modeled with this language and that there is a complete and sound calculus for this language.

1.1 Syntax

First we solidify the syntactic structure of terms.

Definition 3.1 Let V be a set of variables, K a set of constants, and F a set of function symbols. The sets V , K and F are pairwise disjoint. We define the set of terms recursively:

- All variables and constants are (atomic) terms.
- If $t_1, \dots, t_n$ are terms and f an n -place function symbol, then $f(t_1, \dots, t_n)$ is also a term.

Some examples of terms are $f(\sin(\ln(3)), \exp(x))$ and $g(g(g(x)))$. To be able to establish logical relationships between terms, we build formulas from terms.

Definition 3.2 Let P be a set of predicate symbols. Predicate logic formulas are built as follows:

- If $t_1, \dots, t_n$ are terms and p an n -place predicate symbol, then $p(t_1, \dots, t_n)$ is an (atomic) formula.
- If A and B are formulas, then $\neg A$, (A) , $A \wedge B$, $A \vee B$, $A \Rightarrow B$, $A \Leftrightarrow B$ are also formulas.
- If x is a variable and A a formula, then $\forall x A$ and $\exists x A$ are also formulas. $\forall$ is the universal quantifier and $\exists$ the existential quantifier.
- $p(t_1, \dots, t_n)$ and $\neg p(t_1, \dots, t_n)$ are called literals.

Table 3.1 Examples of formulas in first-order predicate logic. Please note that mother here is a function symbol

Formula	Description
$\forall x (\text{frog}(x) \Rightarrow \text{green}(x))$	All frogs are green
$\forall x ((\text{frog}(x) \wedge \text{brown}(x)) \Rightarrow \text{big}(x))$	All brown frogs are big
$\forall x \text{ likes}(x, \text{cake})$	Everyone likes cake
$\neg \forall x \text{ likes}(x, \text{cake})$	Not everyone likes cake
$\neg \exists x \text{ likes}(x, \text{cake})$	No one likes cake
$\exists x \forall y \text{ likes}(y, x)$	There is something that everyone likes
$\exists x \forall y \text{ likes}(x, y)$	There is someone who likes everything
$\forall x \exists y \text{ likes}(y, x)$	Everything is loved by someone
$\forall x \exists y \text{ likes}(x, y)$	Everyone likes something
$\forall x (\text{customer}(x) \Rightarrow \text{likes}(\text{bob}, x))$	Bob likes every customer
$\exists x (\text{customer}(x) \wedge \text{likes}(x, \text{bob}))$	There is a customer whom bob likes
$\exists x (\text{baker}(x) \wedge \forall y (\text{customer}(y) \Rightarrow \text{likes}(x, y)))$	There is a baker who likes all of his customers
$\forall x \text{ older}(\text{mother}(x), x)$	Every mother is older than her child
$\forall x \text{ older}(\text{mother}(\text{mother}(x)), x)$	Every grandmother is older than her daughter's child
$\forall x \forall y \forall z ((\text{rel}(x, y) \wedge \text{rel}(y, z)) \Rightarrow \text{rel}(x, z))$	rel is a transitive relation

- Formulas in which every variable is in the scope of a quantifier are called first-order sentences or closed formulas. Variables which are not in the scope of a quantifier are called free variables.
- Definitions 2.8 (CNF) and 2.10 (Horn clauses) hold for formulas of predicate logic literals analogously.

In Table 3.1 several examples of PL1 formulas are given along with their intuitive interpretations.

1.2 Semantics

In propositional logic, every variable is directly assigned a truth value by an interpretation. In predicate logic, the meaning of formulas is recursively defined over the construction of the formula, in that we first assign constants, variables, and function symbols to objects in the real world.

Definition 3.3 An interpretation $\mathcal{I}$ is defined as

- A mapping from the set of constants and variables $K \cup V$ to a set W of names of objects in the world.
- A mapping from the set of function symbols to the set of functions in the world. Every n -place function symbol is assigned an n -place function.
- A mapping from the set of predicate symbols to the set of relations in the world. Every n -place predicate symbol is assigned an n -place relation.

Example 3.1 Let c_1, c_2, c_3 be constants, "plus" a two-place function symbol, and "gr" a two-place predicate symbol. The truth of the formula

$$F \equiv \text{gr}(\text{plus}(c_1, c_3), c_2)$$

depends on the interpretation $\mathbb{I}$. We first choose the following obvious interpretation of constants, the function, and of the predicates in the natural numbers:

$$\mathbb{I}_1 : c_1 \mapsto 1, c_2 \mapsto 2, c_3 \mapsto 3, \quad \text{plus} \mapsto +, \quad \text{gr} \mapsto >.$$

Thus the formula is mapped to

$$1 + 3 > 2, \quad \text{or after evaluation} \quad 4 > 2.$$

The greater-than relation on the set $\{1, 2, 3, 4\}$ is the set of all pairs (x, y) of numbers with $x > y$, meaning the set $G = \{(4, 3), (4, 2), (4, 1), (3, 2), (3, 1), (2, 1)\}$. Because $(4, 2) \in G$, the formula F is true under the interpretation $\mathbb{I}_1$. However, if we choose the interpretation

$$\mathbb{I}_2 : c_1 \mapsto 2, c_2 \mapsto 3, c_3 \mapsto 1, \quad \text{plus} \mapsto -, \quad \text{gr} \mapsto >,$$

we obtain

$$2 - 1 > 3, \quad \text{or } 1 > 3.$$

The pair $(1, 3)$ is not a member of G . The formula F is false under the interpretation $\mathbb{I}_2$. Obviously, the truth of a formula in PL1 depends on the interpretation. Now, after this preview, we define truth.

1.3 Definition

- An atomic formula $p(t_1, \dots, t_n)$ is true (or valid) under the interpretation $\mathbb{I}$ if, after interpretation and evaluation of all terms $t_1, \dots, t_n$ and interpretation of the predicate p through the n -place relation r , it holds that

$$(\mathbb{I}(t_1), \dots, \mathbb{I}(t_n)) \in r.$$

- The truth of quantifierless formulas follows from the truth of atomic formulas—as in propositional calculus—through the semantics of the logical operators defined in Table 2.1 on page 25.
- A formula $\forall x F$ is true under the interpretation $\mathbb{I}$ exactly when it is true given an arbitrary change of the interpretation for the variable x (and only for x)
- A formula $\exists x F$ is true under the interpretation $\mathbb{I}$ exactly when there is an interpretation for x which makes the formula true.

The definitions of semantic equivalence of formulas, for the concepts satisfiable, true, unsatisfiable, and model, along with semantic entailment (Definitions 2.4, 2.5, 2.6) carry over unchanged from propositional calculus to predicate logic.